

Reviewer Pack — Minimal Rank of Tropicalized Hodge Decomposition vs BP_ReadT...

Entry 0cdfb348d91a · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Rank of Tropicalized Hodge Decomposition vs BP_ReadTwice Width

Entry ID: 0cdfb348d91a **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-23 09:04:04 UTC

1. Conjecture

Field A (mathematical branch): Algebraic Geometry (Hodge Decomposition) **Field B** (complexity object): Complexity Theory: BP_ReadTwice Width

Statement:

[‘For every CNF instance I , the minimal rank of the tropicalized Hodge decomposition of the corresponding algebraic variety $V(I)$ is upper-bounded by a function $f(n)$ such that $\rho(V(I)) = O(f(n))$ and for the trivial BP computing inner product modulo 2, $\rho(V(IP_2)) = \Omega(n)$.’, ‘Equivalently, for all CNF instances I , there exists an algorithm A that computes the BP_ReadTwice width of the instance in time $O(f(n))$.’]

Rationale (proposer’s reasoning):

[‘The Hodge decomposition provides a way to analyze the geometry of algebraic varieties. If this structure can be related to the complexity of BP_ReadTwice, it might reveal new insights into the complexity theory.’, ‘Tropicalization is a method that simplifies complex geometric objects and might help in understanding their relationship with computational problems.’]

Taxonomy category: BP_READTWICE (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 474e83efa44d0ca8

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The conjecture is supported if the computed minimal rank $\rho(V(I))$ is within $O(f(n))$ and BP_ReadTwice width is within $O(f(n))$, for at least 80% of seeds, with a mean difference between $\rho(V(I))$ and $f(n) \leq 3$ and no seed showing BP_ReadTwice width > 10 .

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: ADJACENT_OK against 6 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - "tropicalized Hodge decomposition" AND "BP_ReadTwice Width" - "minimal rank" IN ALGEBRAIC GEOMETRY AND BP_READTWICE WIDTH" - "algorithm for BP_ReadTwice width" related to TROPICALIZED HODGE DECOMPOSITION

Top relevant hits considered: - [http://arxiv.org/abs/0912.2255v3] Test ideals via algebras of p^{-e} -linear maps - [http://arxiv.org/abs/1812.11710v1] Geometric Satake correspondence for affine Kac-Moody Lie algebras of type A - [http://arxiv.org/abs/1705.03356v2] Growth in varieties of multiplier algebras and Groebner bases in operads - [http://arxiv.org/abs/1709.08318v2] Hodge decomposition and the Shapley value of a cooperative game - [http://arxiv.org/abs/2605.13834v1] Topology-Preserving Neural Operator Learning via Hodge Decomposition - [http://arxiv.org/abs/1807.06283v2] Tropical Fano Schemes

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=1, elapsed=0.1s

5.1 Generated Python source

```
import random
import math
from fractions import Fraction

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def gaussian_elimination(A):
        m, n = len(A), len(A[0])
        for i in range(m):
            max_row = i
            for j in range(i+1, m):
                if abs(A[j][i]) > abs(A[max_row][i]):
                    max_row = j
            A[i], A[max_row] = A[max_row], A[i]
            pivot = A[i][i]
            for j in range(n):
                A[i][j] /= pivot
            for j in range(m):
                if j != i:
                    factor = A[j][i]
                    for k in range(n):
```

```

        A[j][k] -= factor * A[i][k]

    return A

def matrix_multiplication(A, B):
    m, n, p = len(A), len(B), len(B[0])
    C = [[0] * p for _ in range(m)]
    for i in range(m):
        for j in range(p):
            for k in range(n):
                C[i][j] += A[i][k] * B[k][j]
    return C

def bp_readtwice_width(instance):
    # Placeholder function to simulate BP_ReadTwice width computation
    n = len(instance)
    return 2 * n # Simplified for demonstration purposes

def tropicalized_hodge_rank(instance):
    # Placeholder function to simulate tropicalized Hodge rank computation
    n = len(instance)
    A = [[random.randint(0, 1) for _ in range(n)] for _ in range(n)]
    rank = gaussian_elimination(A)
    return sum(1 for row in rank if any(row[j] != 0 for j in range(len(row))))

def f_n(n):
    # Placeholder function to simulate the upper bound function f(n)
    return n * math.log2(n)

instance = [[random.randint(0, 1) for _ in range(random.randint(5, 10))] for _ in range(random.randint(5, 10))]
rho_V_I = tropicalized_hodge_rank(instance)
bp_width = bp_readtwice_width(instance)
f_n_value = f_n(len(instance))

conjecture_holds = (rho_V_I <= f_n_value) and (bp_width <= 10)

return {
    "metric_name": "BP_ReadTwice Width",
    "metric_value": bp_width,
    "instances_tested": 1,
    "conjecture_holds": conjecture_holds,
    "counterexample": f"rho_V_I={rho_V_I}, f(n)={f_n_value}, bp_width={bp_width}" if not conjecture_holds
}

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] or [2, 3, 5, 7, 11, 13, 17, 19, 23, 29] * 3

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    mean_value = sum(r["metric_value"] for r in results) / len(results)
    std_value = math.sqrt(sum((r["metric_value"] - mean_value) ** 2 for r in results) / len(results))
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

```

```

if all(r["conjecture_holds"] for r in results):
    print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_fraction}")
elif support_fraction >= 0.8:
    print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_fraction}")
else:
    first_failing_seed = next((r["seed"] for r in results if not r["conjecture_holds"]), None)
    counterexample = next(r["counterexample"] for r in results if not r["conjecture_holds"])
    print(f"RESULT: FALSIFIED counterexample=\"{counterexample}\" first_failing_seed={first_failing_seed}")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```

TRIAL: {'metric_name': 'BP_ReadTwice Width', 'metric_value': 16, 'instances_tested': 1, 'conjecture_holds': True}
TRIAL: {'metric_name': 'BP_ReadTwice Width', 'metric_value': 14, 'instances_tested': 1, 'conjecture_holds': True}

```

--STDERR--

Traceback (most recent call last):

```

File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_0d63bd1e.py", line 85, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^

```

```

File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_0d63bd1e.py", line 65, in run_trial
    rho_V_I = tropicalized_hodge_rank(instance)
             ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^

```

```

File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_0d63bd1e.py", line 57, in tropicalized_hodge_rank
    rank = gaussian_elimination(A)
          ^^^^^^^^^^^^^^^^^^^^^

```

```

File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_0d63bd1e.py", line 31, in gaussian_elimination
    A[i][j] /= pivot

```

ZeroDivisionError: float division by zero

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test code crashed before producing data, which prevents us from evaluating whether the conjecture holds or not. | next: Investigate and fix the crash in the test code to proceed with the verification of the conjecture.

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	15319
2	preregistration	ollama_remote	glm4:latest	0	0	9626
3	novelty	ollama_remote	glm4:latest	0	0	8474
4	novelty	ollama_remote	glm4:latest	0	0	10525
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	14961
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	11690
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	7689
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	10989
9	judge	ollama_remote	glm4:latest	0	0	11451

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 100725 ms total latency. Provider mix: {'ollama_remote': 9}

(full prompt+response transcripts available in `research/audit/0cdfb348d91a.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/0cdfb348d91a.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/0cdfb348d91a.tar.gz` (if generated)